

Nutrition and loading to improve fascia function

Danielle Steffen and Keith Baar

Introduction

Connective tissues, including bone, cartilage, tendon, ligaments, and fascia, are critical components of a functioning musculoskeletal system. These tissues serve to absorb, dissipate, and transmit the forces applied to the body during movement. Each one is specialized for one or more mechanical function based on the location in the body and the resulting amount and type of force applied to the cells within the tissue. Tendons, ligaments and fascia function primarily to transfer force within the musculoskeletal system, and fascia has the added function of facilitating movements by lubricating the interface between tissues. To facilitate the transfer of force, these connective tissues are composed of an extracellular matrix (ECM), structural proteins that give them a high tensile strength (Kjaer, 2004), and ground substance that brings water into the tissue to provide compressive strength. The primary structural proteins are collagen, laminin, and elastin, with collagen contributing ~80% of the total protein. There are 28 different collagen proteins (Ricard-Blum, 2011). Within the musculoskeletal connective tissues, type I collagen makes up more than 90% of the collagen fraction, with type III collagen comprising much of the remaining fraction. Therefore, types I and III collagen are the major fibrillar collagens that provide tensile strength to the musculoskeletal connective tissues (Kjaer, 2004).

To increase the strength of the tissues, the proteins that form the matrix are cross-linked (Figure 6.1). Cross-linking can occur in two ways: enzymatically and non-enzymatically. Enzymatic cross-links are mediated primarily by lysyl oxidase (LOX; 26, 44, 45), a copper-dependent enzyme that forms intermolecular bonds between lysine residues at the end of collagen molecules. By contrast, non-enzymatic cross-links are formed through a Maillard reaction between a sugar and an amino acid at any location within the matrix. These cross-links are called advanced glycation end-products (AGE), and, as would be expected, are higher in diabetics (Dyer et al., 1993) where blood glucose levels are chronically increased. Both enzymatic cross-linking, through LOX, and non-enzymatic cross-linking through AGEs, increase the stiffness of the matrix (Reddy, Stehno-Bittel and Enwemeka, 2002; Marturano et al., 2013). The main differences between enzymatic and non-enzymatic cross-links are their location and turnover rate, where AGEs can be made throughout the matrix and this decreases collagen turnover and over time impairs matrix function (Hammes et al., 1991; Corman et al., 1998). Further, when AGEs are high and collagen levels are low (for example in individuals who are inactive), the matrix becomes brittle (Akeson et al., 1977; Calve et al., 2005), resulting in an increase in the likelihood of catastrophic injuries in sport (Myer et al., 2011).

Chapter 6

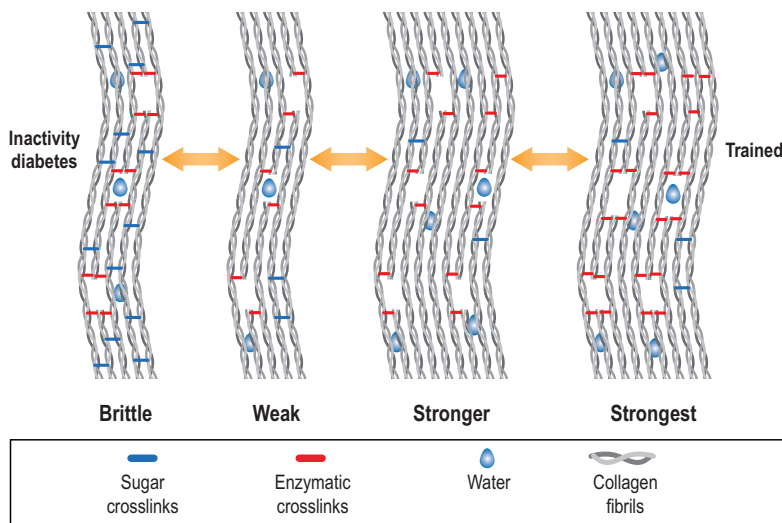


Figure 6.1

Determinants of musculoskeletal soft tissue stiffness. The major determinants of soft tissue stiffness are depicted in this cartoon. Collagen (gray) provides the mechanical backbone for the tissues; the more collagen fibers, the more robust the tissue. The collagen matrix can be stiffened by either enzymatic (red) or sugar based (advanced glycation endproducts; blue) cross-links. Water (blue drops) held within the matrix makes the tissue viscoelastic. A tissue becomes stiffer and stronger as both collagen and enzymatic cross-links increase. In contrast, a tissue with little collagen and a lot of sugar cross-links is stiff but not strong (brittle).

The major proteins of the ground substance are proteoglycans (PGs), proteins that have highly charged glycosaminoglycan (GAG) side chains. These charged GAGs serve to modulate tissue hydration and/or help organize the collagen matrix (Sasarman et al., 2016; Robinson et al., 2017). The water content of the musculoskeletal tissues is determined by the retention of free water by PGs within the matrix (Dubinskaya et al., 2007). Larger PGs are more common in compressed musculoskeletal tissues (Evanko and Vogel, 1990) and provide compressive strength by bringing large amounts of water into the tissue. By contrast, smaller PGs (biglycan and decorin) are more prevalent in tissues loaded in tension (Robinson et al., 2017). Compressive strength is utilized both during tensile loading (as the tissue is stretched), when the water within the tissue resists radial compression as the collagen molecules are squeezed together, and during compressive loading, such as when tissues wrap around bones or retinacula. Together, the dense

cross-linked collagen matrix and proteoglycan-rich ground substance provide the mechanical strength of the tissues and therefore specific connective tissues use different ratios of these substances to meet their specific mechanical requirements.

In contrast to the classical view of the musculoskeletal connective tissues being relatively inert, stable tissues where the collagen has a half-life of >105 years (Heinemeier et al., 2013), more recent data suggests that connective tissues are highly dynamic. Smeets and colleagues (Smeets et al., 2019) gave subjects undergoing knee replacement surgery phenylalanine containing a heavy isotope of carbon two hours before and then throughout the operation. During surgery, all the tissues of the knee were collected and the incorporation of heavy phenylalanine into the proteins contained in those tissues was determined. As a control, the authors took a muscle biopsy at the time of surgery so that they could compare the tissues of the knee to a highly dynamic tissue.

Nutrition and loading to improve fascia function

In contrast to expectation, protein turnover in the anterior cruciate ligament (ACL) and patellar tendon was greater than the muscle tissue, cartilage protein turnover was similar to muscle, and synovial tissue turned over almost 3-times faster than muscle (Smeets et al., 2019). In support of the findings of Smeets and colleagues, Myrick et al recently showed that, over the course of an 8-month soccer season, the ACL of young women increased in size, suggesting that there was an increase in collagen in the tissue (Myrick et al., 2019). Together, these data suggest that the musculoskeletal connective tissues are highly dynamic in their response to load.

The primary forces within the musculoskeletal system are *tension*, i.e. when the Achilles tendon is loaded in one direction during plantar flexion (Figure 6.2A); *compression*, i.e. when the meniscus absorbs the impact force of a step (Figure 6.2B); and *shear*, i.e. when the superior extensor retinaculum allows the tendons of the

tibialis anterior slide beneath it during dorsi flexion (Figure 6.2C). When the musculoskeletal system is working properly, the specific load placed on the tissue results in a stimulus that orchestrates a cascade of events within the cell that results in changes to the proteins that make up the ECM. This is most notable in embryonic development because compressive loads induce fibrocartilage formation and tensional loads produce an aligned collagen matrix within a tendon (Evanko and Vogel, 1990; Kapacee et al., 2008). This molecular response is critical to the growth, differentiation and development of the cells and dictates the resulting function of the tissue.

The connective tissues are largely made from a population of fibroblast cells that contain the high mobility group (HMG) box transcription factor SOX9 (Asou et al., 2002). Even though the cells have a similar origin, once the cells migrate into position and begin being loaded, they

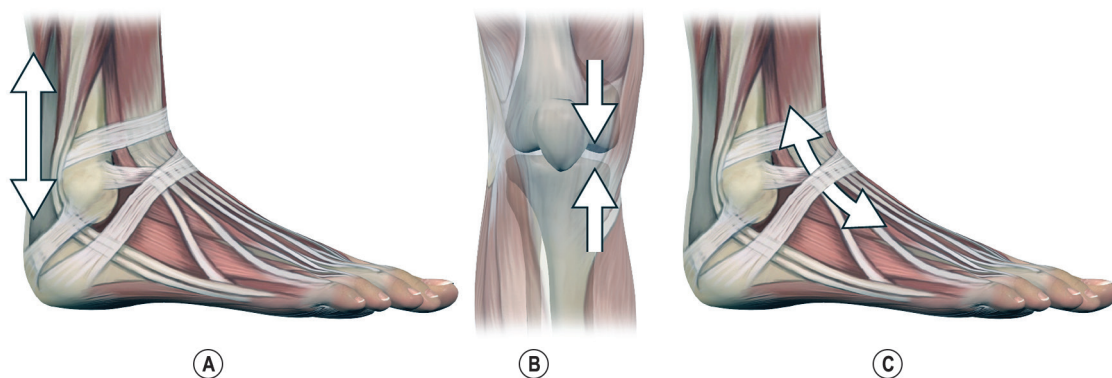


Figure 6.2

Loads within the musculoskeletal system. The three major loads in the musculoskeletal system are: **(A) Tension**, when a tissue is pulled in one direction; **(B) Compression**, when a tissue is pushed in on itself; and **(C) Shear**, when two tissues slide past each other.

Chapter 6

differentiate to fit their mechanical environment. In fact, fibroblast cells isolated from connective tissues at different locations in the body become transcriptionally distinct cell types (Chang et al., 2002). These data suggest that the mechanical environment is essential to driving cell fate. Therefore, understanding the mechanical environment, and the cellular response to load, is essential for maximizing musculoskeletal cell and tissue function.

Beyond the mechanical environment, connective tissues are dependent on nutritional cues. As stated above, the primary protein within connective tissues is collagen, the most abundant protein in the body. In order for the collagen made in the cells to be released into the ECM, it has to be modified by the enzyme prolyl hydroxylase (Mussini, Hutton and Udenfriend, 1967). During the hydroxylation reaction, prolyl hydroxylase consumes vitamin C. Since humans lack the enzyme L-gulonolactone oxidase (Nishikimi et al., 1988), we are unable to make vitamin C and therefore we must consume enough of this micronutrient to support normal collagen turnover. While this is the most direct nutritional input needed for healthy connective tissues, other foods are suggested to be important in connective tissue function. These foods and how they interact with loading to promote tendon, ligament and fascial structure and function will be the focus of this chapter.

Cellular and molecular response to loading

Tendons, ligaments, and fascia are formed by a dense and aligned collagen matrix. While these three tissues are structurally similar, they can undergo regional specialization based on their mechanical environment. For example, retinacular fascial tissues, that need to lubricate

the constant sliding of underlying tendons, have more than 10-fold higher levels of hyaluronan, a glycosaminoglycan (GAG) that functions as a lubricant, than the epimysial fascia, that surrounds muscles where shear forces are lower (Fede et al., 2018). Similarly, the flexor digitorum profundus tendon of rabbits is structurally and functionally two distinct tendons; the tendon proper, that is loaded in tension, and a fibrocartilage portion, where the tendon is compressed as it passes through the carpal tunnel. Even though the cells of the two regions have the same origin, the total GAG content of the tendon varies from 0.2% of the dry weight in the tendon, up to 6.5% as the tissue takes on a fibrocartilage phenotype (Gillard et al., 1979). When the compressed region is experimentally released from the carpal tunnel, the GAG content within the tendon rapidly drops from 6.5 to 2.5% after only eight days and further decreases to below 1% within six months after the release surgery (Gillard et al., 1979). To determine whether compressive load was sufficient to induce a fibrocartilage phenotype, Robbins and colleagues removed deep flexor tendons from cows, subjected them to 72 hours of cyclic compressive load, and then measured markers of cartilage gene expression. Just three days of compressive loading was enough the increase expression of cartilage enriched large PGs, such as aggrecan up to 4.5-fold (Robbins, Evanko and Vogel, 1997). Therefore, in tendon compressive loading is sufficient to induce and maintain a fibrocartilage phenotype.

The type of load also influences the size and function of the PGs present within a tendon. Fibrocartilage present in compressed tendon regions is typified by larger PG. These larger PG increase water within the tissue to increase compressive strength. Smaller PGs are enriched in embryonic tendons and adult tendons under tension (Evanko and Vogel, 1990; Ehlers and Vogel,

Nutrition and loading to improve fascia function

1998). As the developmental tendon is compressed, epigenetic specialization begins and the cells within the compressed region take on a fibrocartilage phenotype, that can be recreated using *in vitro* cyclic compression (Evanko and Vogel, 1993). In the region of the tendon undergoing tensile loading, smaller PGs have a role in collagen fibrillogenesis—defining the outer surface of individual collagen fibrils. When these smaller PGs are missing, the size of the fibrils becomes disrupted and the mechanical integrity of the tendon is lost (Ameye and Young, 2002; Robinson et al., 2017).

Unlike PGs and GAGs, lubricating proteins such as Prg4/lubricin are regulated by shear stress (Nugent et al., 2006). Simply applying a shear stress to cells for 24 hours results in a 3-fold increase in lubricin secretion. Since knocking

out lubricin makes tendons stick together more tightly (Hayashi et al., 2013), this suggests that shear forces are necessary to maintain the lubrication between tendons/fascial layers. Together, these data suggest that compressive and shear loads rapidly increase the expression of genes involved in cartilage formation and lubrication, respectively (Figure 6.3B, C).

Where compression and shear increase larger PGs and lubrication, tensional forces are necessary for the production of an aligned collagen matrix (Kapacee et al., 2008). The requirement for tension in the synthesis of a robust and aligned matrix is best seen when the load is removed. For example, when Majima and colleagues inserted a stainless steel wire between the tibial plateau and the base of the patella to release tension from the patellar tendon, nuclei per mm² increased

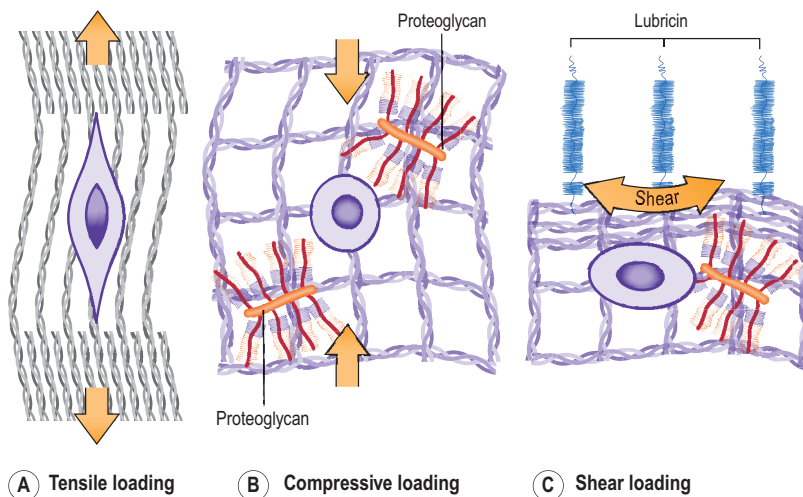


Figure 6.3

Different loads result in different cellular responses. In response to (A) tensile loading, cells secrete type I collagen and make lysyl oxidase resulting in a stiff aligned collagen matrix. In contrast, (B) compressive loads on the same cells make proteoglycans that contain a protein like hyaluronic acid and glycosaminoglycans like chondroitin and keratan sulfate. Over time this results in a decrease in collagen alignment and an increase in water in the tissue (held by the GAGs). Lastly, (C) shear loads lead to the production of proteoglycans and lubricin at the edge of the tissue, resulting in a collagen matrix that contains high amounts of fluid at the edge of the tissue to lubricate movement.

Chapter 6

~7-fold, collagen fibril diameter decreased ~40%, and collagen bundles became less well-oriented (Majima et al., 2003). These data indicate that without tensional load healthy connective tissues take on a scarred phenotype, becoming more cellular with smaller, less aligned collagen fibrils. By contrast, one hour of kicking exercise resulted in a dynamic increase in patellar tendon collagen synthesis (Miller et al., 2005). Collagen synthesis, as measured using stable isotope infusion followed by tendon biopsies, was elevated ~40% 6 hours after the exercise bout, peaked at 80% greater 24 hours after the exercise, and was still 40% higher 72 hours after (Miller et al., 2005). Interestingly, the same group found that the one hour kicking exercise did not increase collagen synthesis in women (Miller et al., 2007); however, this could be a result of the time point of measurement (72 hours in the women), 48 hours after the peak of collagen synthesis in men (Miller et al., 2005). An even greater increase in collagen synthesis is seen in muscle with tensile loading where collagen synthesis increases ~185% 6 hours after exercise, stays at that level 18 hours later, before returning to baseline 72 hours after finishing the exercise bout (Miller et al., 2005). In chicken muscle, tensional loading increases collagen protein synthesis 5-fold, resulting in a 10-fold increase in the amount of collagen within the muscle after as little as two days (Laurent, McAnulty and Gibson, 1985). These data suggest that tensile loading increases collagen synthesis, resulting in an increase in collagen within connective tissues (Figure 6.3A).

In order to understand what about tensile loading drives an increase in collagen synthesis, we turned to an *in vitro* model of a ligament (Paxton, Grover and Baar, 2010). Three dimensional (3D) engineered ligaments are structurally and functionally very similar to embryonic ligaments. Using these cultured ligaments and a molecular

marker that is necessary for the load-induced increase in collagen synthesis (the extracellular-regulated kinase; ERK; 22), we characterized what aspects of tensile loading connective tissue cells transduce into a signal that results in more collagen production (Paxton et al., 2012). These experiments demonstrated that the frequency and amplitude of the stretch on a ligament had similar effects on the cellular response, even very small loads fully activated ERK. In contrast, the duration of an exercise bout was very important; the cellular response to tensile loading peaked between five and ten minutes and then returned to baseline levels by 90 minutes (Figure 6.4A).

These data suggested that the cells were quickly becoming refractory to the tensile load. To determine how long it took to regain the capacity to signal, ligaments were loaded for ten minutes and then rested for 0.5, 1, 3, 6, and 12 hours (Figure 6.4B). Full signaling returned to the ligament cells only after resting for six hours (Paxton et al., 2012). Lastly, an intermittent tensile loading program (4 x 10 minutes separated by six hours) was compared with a continuous loading program, and the ligaments that received 40 minutes of tensile load each day in short bouts increased collagen synthesis twice as much as those being loaded 24 hours a day. These data are similar to the response in bone where cells become refractory after ~40 loads and take 6-8 hours to regain the ability to signal (Burr, Robling and Turner, 2002). Together, these data suggest that musculoskeletal connective tissues increase collagen synthesis in response to short periods (~10 minutes) of tensile load separated by load periods (~6 hours) of rest (Figure 6.4C).

Nutrition for connective tissues

As discussed above, the structure and function of tendons, ligaments, and fascia can be directly

Nutrition and loading to improve fascia function

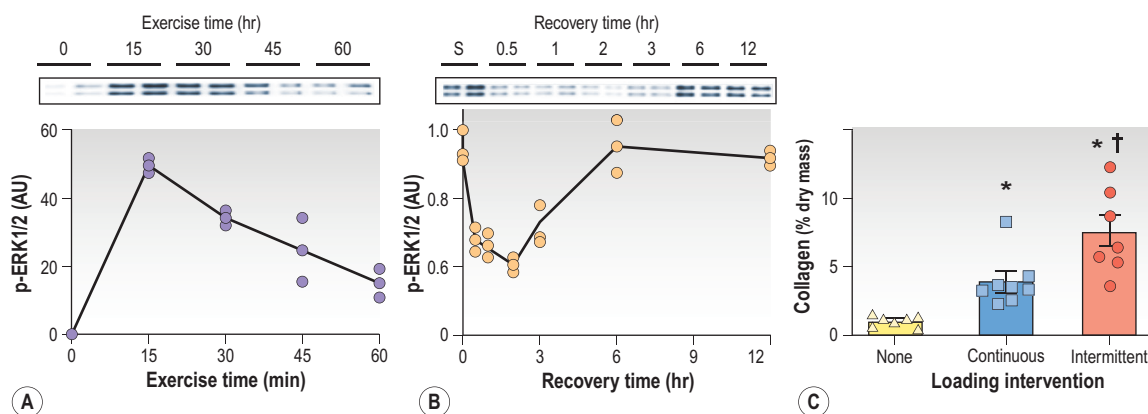


Figure 6.4 Short loads and long rest periods are optimal for musculoskeletal connective tissues.

Data are presented from engineered ligaments undergoing load (Paxton *et al.*, 2012). **(A)** The cellular response to loading (activation of the Extracellular Regulated Kinase) peaks at 10 minutes of loading (not shown) and progressively decreases even though the tissue is still being loaded. In **(B)**, ligaments were stretched for 10 minutes (S/O time) and then rested for 0.5, 1, 3, 6, or 12 hours before another 10-minute stretch and ERK phosphorylation was determined. The data show that it takes 6 hours of rest before the cells can respond completely to a second bout of load. **(C)** Ligaments were stretched either continuously (24 hours a day) or intermittently (10 minutes of stretch, six hours of rest, 10 minutes of stretch, ...) for five days and the collagen content of the ligaments was determined. Continuous loading increased collagen ~4-fold, whereas an intermittent loading protocol increased collagen ~7.5-fold. This indicates that short bouts of loading (~10 minutes) with long periods of rest (≥ 6 hours) is optimal for building musculoskeletal connective tissues.

affected by nutrition. In the sections below, specific nutrients are discussed vis a vis their ability to improve collagen synthesis.

Vitamin C

Vitamin C is a water-soluble vitamin, meaning that we need to consume an adequate amount every day. In industrialized societies, daily consumption of the 46mg per day is not difficult for most people. This was not the case 250 years ago when seamen in the British Navy on long voyages would routinely get scurvy, a disease characterized by easy bruising, tooth loss, the opening of old wounds, and the stiffening of tendons. In the first controlled nutrition study ever reported, the Scottish physician James Lind fed 12 sailors with scurvy one of six different interventions (Lind, 1753). The two sailors who ate two oranges and one lemon each recovered completely within

a week, whereas in the other sailors the symptoms continued. It took another 200 years before Jerome Gross showed that guinea pigs on a vitamin C deficient diet did not synthesize collagen (Gross, 1959), finally making the molecular connection between vitamin C, collagen synthesis and scurvy. Vitamin C is consumed by the enzyme prolyl hydroxylase in the first post-translational reaction following the synthesis of collagen. Without vitamin C, prolyl hydroxylase is inactivated and the collagen triple helices become unstable and can no longer be exported (Mussini, Hutton and Udenfriend, 1967). Since vitamin C is consumed in the hydroxylation reaction, it must be replaced every day. This also means that vitamin C levels are low in the morning and need to be replenished before the body can synthesize collagen in response to loading (Lis and Baar, 2019). Even though the

Chapter 6

requirement for vitamin C in the synthesis of collagen has been recognized for 250 years, whether adding extra vitamin C increases collagen synthesis has yet to be determined.

Glycine

The proteins within tendons, ligaments and fascia contain ~40% glycine by weight (Smeets et al., 2019). Because of the amount of glycine in these tissues, some researchers have hypothesized that increasing dietary glycine would benefit tendon healing. To test this hypothesis, Vieira and colleagues injured the Achilles tendon of rats using collagenase and then fed half of the rats a diet containing 5% glycine. After three weeks, the rats on the 5% glycine diet had greater collagen and glycosaminoglycan content than those on the control diet (C. P. edroz. Vieira et al., 2015). This increase in collagen and GAGs resulted in improved mechanical strength both in this study and the follow up study where they added green tea to the nutritional supplement (C. P. edroz. Vieira et al., 2015; C. P. Vieira et al., 2015), suggesting that glycine may aide in the recovery of tendon function after injury. It is important to note that these studies have yet to be repeated in human subjects and that most people would find it difficult to consume ~25 grams of glycine a day. This is equivalent to a quarter of the bottle of most popular glycine supplements. However, given the success of the rodent data, human studies in this area are warranted.

Gelatin/hydrolyzed collagen

A natural source of glycine is dietary collagen. As of 2019, dietary collagen is the fastest growing supplement, with sales topping \$45 million in the United States in 2018. As described above, collagen is ~35% glycine and therefore consuming collagen could improve collagen synthesis within the connective tissues. The two most popular

ways to consume collagen in the diet are gelatin and hydrolyzed collagen. Gelatin is released from the skin, bones, and other connective tissues of animals such as cows, pig, and fish as a result of boiling. As a result, bone broths, soups made by boiling an animal carcass, are rich in gelatin; however, the amount of gelatin in a given bone broth varies from 432–9,000 mg per serving depending on how it is made (Alcock, Shaw and Burke, 2019). For more consistent gelatin levels, commercial products from Jello® to Knox® gelatin can be found in most grocery stores. These products are normally used to make desserts since dissolving gelatin in hot water and then cooling it for ~4 hours results in the formation of jelly. For those looking to dissolve dietary collagen in a liquid, hydrolyzed collagen dissolves better than gelatin. For hydrolyzed collagen, gelatin is further processed using either acid or enzymes such as pepsin, papain, bromelain and protease (Hema et al., 2017). These proteins are proteases, enzymes that cut proteins into smaller pieces called peptides. Because processing with acids or enzymes cuts the gelatin into smaller pieces (called peptides), hydrolyzed collagen is unable to form a gel and is more soluble in room temperature water. Even though different companies use different processes that result in different collagen peptides, whether this has any biological significance has yet to be determined.

To determine whether dietary collagen could increase collagen synthesis in connective tissues, we performed a randomized double-blinded placebo-controlled crossover designed study on subjects who consumed 0, 5, or 15 grams of gelatin (Shaw et al., 2017). Over a three-day period, young men took a supplement containing 48mg of vitamin C and either a placebo, 5, or 15 grams of gelatin. The subjects took the supplement every six hours between 7am and 7pm. One hour after taking the supplement, the subjects jumped

Nutrition and loading to improve fascia function

rope to stimulate bone collagen synthesis. Interestingly, consuming 15 grams of gelatin doubled the amount of collagen synthesized in either the placebo or 5 gram groups (Shaw et al., 2017). Furthermore, when blood was taken from the subjects an hour after taking the supplement and the serum was added to engineered ligaments, the engineered ligaments increased their collagen content in a dose-dependent manner. These data suggest that dietary collagen together with sufficient vitamin C increases collagen synthesis in tendons and ligaments. Repeating the study with gelatin, hydrolyzed collagen, or a gummy made with half gelatin and half hydrolyzed collagen demonstrated that the gelatin and hydrolyzed collagen increased collagen synthesis equivalently, whereas the gummy had no effect (Lis and Baar, 2019). The lack of an effect of the gummy is likely the result of boiling the juice and killing the vitamin C, since amino acid content in the blood was similar between the three feeding groups. Since dietary collagen can increase musculoskeletal collagen synthesis, it is not surprising that consumption of 10g of hydrolyzed collagen decreased knee pain from standing and walking in a randomized double-blinded placebo-controlled study in athletes (Clark et al., 2008). The decrease in knee pain could be the result of improved collagen synthesis in cartilage since cartilage thickness, measured using gadolinium labeled MRI, increases with long term consumption of hydrolyzed collagen (McAlindon et al., 2011). In support of this work, a prospective double-blinded placebo-controlled clinical trial with a cross-over design showed that runners experiencing Achilles pain benefited from collagen supplementation (Praet et al., 2019). In the study, injured runners were placed on an exercise rehabilitation program with or without supplementation with hydrolyzed collagen. Those runners in the collagen supplementation group

had a significantly better outcome than those in the placebo group (Praet et al., 2019). Together, these data suggest that consuming gelatin or hydrolyzed collagen may increase collagen synthesis; however, much of the work is still preliminary and needs more research to support its widespread use.

Other nutrients

Because musculoskeletal sprains, strains, and pulls account for a third of all missed workdays and the only outcome measure in many studies is pain, a myriad of other nutrients are purported to improve tendon/ligament function. These and other nutraceuticals have recently been expertly reviewed by Fusini and colleagues (Fusini et al., 2016) and we would encourage readers to go there for details on most of these agents. Many of the nutrients reviewed by Fusini and colleagues are thought to be anti-inflammatory. These agents may therefore work by decreasing pain rather than affecting the structure and function of the musculoskeletal connective tissues. Therefore, much more work is needed to validate these purported nutraceuticals.

Combining loading and nutrition for connective tissues

Tendons and ligaments have relatively low blood flow, instead using diffusion from the environment to provide nutrients and remove waste products (Skyhar et al., 1985; Amiel et al., 1986). Most oxygen and nutrients come from the myotendinous junction (Manske and Lesker, 1982), the fat pad at the enthesis (Benjamin et al., 2004), the paratenon or the synovial sheath (Manske, Bridwell and Lesker, 1978; Lundborg, Holm and Myrhage, 1980; Whiteside and Sweeney, 1980; Amiel et al., 1986). Similar to a sponge, fluid within connective tissues moves out as

Chapter 6

the tissue is loaded and when the tissue relaxes the fluid flows back into the tissue, increasing nutrient uptake (Lundborg, Holm and Myrhage, 1980). Given that loading of connective tissue increases nutrient delivery, the timing of nutrition relative to load is an important consideration. Therefore, optimal timepoint for exercise following intake of supplement is related to the rate of digestion, adsorption and delivery of the nutrient. For 15g of vitamin C-enriched gelatin, amino acid levels in the blood peak approximately one hour after ingestion (Shaw et al., 2017). The blood isolated from subjects an hour after feeding is sufficient to increase collagen synthesis in engineered ligaments and eating the vitamin C-enriched gelatin an hour before loading results in an increase in collagen synthesis in people (Shaw et al., 2017). These data suggest that loading the target tendons at approximately the peak of appearance in the blood can increase collagen production in musculoskeletal connective tissues.

Three case/limited participant studies have utilized this combined loading and nutrition protocol to improve connective tissue function and return to play. In the first, a professional basketball player with a central core patellar tendinopathy was provided 15g of gelatin in orange juice an hour before a ten-minute exercise protocol (Baar, 2019). This rehabilitation protocol was added on top of his normal training and game schedule and contributed to the complete restoration of patellar tendon structure, as determined by MRI, and elimination of pain. In the second, nutrition and short periods of loading were used to accelerate return to play in two elite rugby players following ACL rupture (Shaw, Serpell and Baar, 2019). Soon after reconstructive surgery, the athletes performed two to three range of motion/loading sessions per day, separated by six hours, and preceded by ingestion of

10 g gelatin with 250 mg vitamin C. The rehab sessions used the idea of a minimal effect dose of load (Paxton et al., 2012) to maximally active collagen synthesis within the newly grafted ligament, and to repair the hamstring tendon that was partially used as the graft, and the timed delivery of nutrients was thought to provide necessary substrates for collagen synthesis (Shaw, Serpell and Baar, 2019). The result was a return to full hamstring strength in under 15 weeks and a return to international rugby in ~30 weeks. The last study using load and nutrition to maximize return to play was the cross-over design study on runners with Achilles pain described above (Praet et al., 2019). In this study, the 19 injured runners all performed 2 × 90 one-legged slow, eccentric drops daily for six months with the leg both straight and bent. Half of the runners were supplemented with a hydrolyzed collagen for the first three months, whereas the others got the placebo. At the halfway point, the groups were switched. In the first three months, twice as many participants returned to running from the hydrolyzed collagen group than the placebo. After the cross-over, the collagen group again had twice as many people return to running. In the end, eight subjects returned to running from the hydrolyzed collagen group, whereas only four returned to running while in the placebo group (Praet et al., 2019).

Conclusions and future focus

There is a clear relationship between load, nutrition and tendon/ligament/fascia function. In response to tension, compression, and shear loads these tissues increase collagen, cartilage, and lubrication gene expression, respectively. The positive effect of loading is best seen in recovery from injury. Starting a loading program within two days after injury decreases return to play time by 25% when compared with a group

Nutrition and loading to improve fascia function

that immobilized the injured area for 9 days (Bayer, Magnusson and Kjaer, 2017). This means that the proper application of load is essential to the development of healthy musculoskeletal connective tissues. When these loads are combined with nutritional interventions such as vitamin C and gelatin or other nutrients like glycine, the result is an improvement in tissue structure and function. Even though there is strong evidence

for how load alters tissue function, the role of nutrition in this process is in its infancy. Larger clinical studies are needed to determine whether dietary collagen has clinical efficacy or is simply the result of small samples sizes and a strong belief effect.

An earlier version of this article originally appeared in Fascia, Function, and Medical Applications (CRC Press 2020) by Lesondak and Akey.

References

- Akeson, W.H. et al. (1977) Collagen Cross-linking alterations in joint contractures: Changes in the reducible cross-links in periarticular connective tissue collagen after nine weeks of immobilization. *Connective Tissue Research*. Informa Healthcare 5(1): 15–19. doi: 10.3109/03008207709152607.
- Alcock, R.D., Shaw, G.C. & Burke, L.M. (2019) Bone broth unlikely to provide reliable concentrations of collagen precursors compared with supplemental sources of collagen used in collagen research. *International Journal of Sport Nutrition and Exercise Metabolism*, 29(3): 265–272. doi: 10.1123/ijsnem.2018-0139.
- Ameys, L. & Young, M.F. (2002) Mice deficient in small leucine-rich proteoglycans: novel in vivo models for osteoporosis, osteoarthritis, Ehlers-Danlos syndrome, muscular dystrophy, and corneal diseases. *Glycobiology*.
- Amiel, D. et al. (1986) Nutrition of cruciate ligament reconstruction by diffusion: Collagen synthesis studied in rabbits. *Acta Orthopaedica Scandinavica*, 57: 201–203.
- Asou, Y. et al. (2002) Coordinated expression of scleraxis and Sox9 genes during embryonic development of tendons and cartilage. *Journal of Orthopaedic Research*. 20: 827–833. doi: 10.1016/S0736-0266(01)00169-3.
- Baar, K. (2019) Stress relaxation and targeted nutrition to treat patellar tendinopathy. *International Journal of Sport Nutrition and Exercise Metabolism*. pp. 1–5. doi: 10.1123/ijsnem.2018-0231.
- Bayer, M.L., Magnusson, S.P. & Kjaer, M. (2017) Early versus delayed rehabilitation after acute muscle injury. *New England Journal of Medicine*. 377(133): 1300–1301. doi: 10.1056/NEJMc1708134.
- Benjamin, M. et al. (2004) Adipose tissue at entheses: The rheumatological implications of its distribution. A potential site of pain and stress dissipation?. *Annals of the Rheumatic Diseases*. 63(12): 1549–1555. doi: 10.1136/ard.2003.019182.
- Burr, D.B., Robling, A.G. & Turner, C.H. (2002) Effects of biomechanical stress on bones in animals. *Bone*. 30(5): 781–786. doi: 10.1016/S8756-3282(02)00707-X.
- Calve, S. et al. (2005) The effect of denervation on the heterogeneous material properties of the tibialis anterior tendon. *Proceedings of the 2005 Summer Bioengineering Conference*.
- Chang, H.Y. et al. (2002) Diversity, topographic differentiation, and positional memory in human fibroblasts. *Proceedings of the National Academy of Sciences of the United States of America*. doi: 10.1073/pnas.162488599.
- Clark, K.L. et al. (2008) 24-Week study on the use of collagen hydrolysate as a dietary supplement in athletes with activity-related joint pain. *Current Medical Research and Opinion*. doi: 10.1185/030079908x291967.
- Corman, B. et al. (1998) Aminoguanidine prevents age-related arterial stiffening and cardiac hypertrophy. *Pharmacology*. Available at: www.pnas.org. (Accessed: 1 July 2020).
- Dubinskaya, V.A. et al. (2007) Comparative study of the state of water in various human tissues. *Eksperimental'noi Biologii i Meditsiny*.
- Dyer, D.G. et al. (no date) Accumulation of Maillard reaction products in skin collagen in diabetes and aging.
- Ehlers, T.W. & Vogel, K.G. (1998) Proteoglycan synthesis by fibroblasts from different regions of bovine tendon cultured in alginate beads. *Comparative Biochemistry and Physiology - A Molecular and Integrative Physiology*. 121(4): 355–363. doi: 10.1016/S1095-6433(98)10144-7.

Chapter 6

- Evanko, S.P. & Vogel, K.G. (1990) Ultrastructure and proteoglycan composition in the developing fibrocartilaginous region of bovine tendon. *Matrix*. 10(6): 420–436. doi: 10.1016/S0934-8832(11)80150-2.
- Evanko, S.P. & Vogel, K.G. (1993) Proteoglycan synthesis in fetal tendon is differentially regulated by cyclic compression *in vitro*. *Archives of Biochemistry and Biophysics*. 307(1): 153–164. doi: 10.1006/abbi.1993.1574.
- Fede, C. et al. (2018) Quantification of hyaluronan in human fasciae: variations with function and anatomical site. *Journal of Anatomy*. 233(4): 552–556. doi: 10.1111/joa.12866.
- Fusini, F. et al. (2016) Nutraceutical supplement in the management of tendinopathies: A systematic review. *Muscles, Ligaments and Tendons Journal*. 6(1): 48–57. doi: 10.11138/mltj/2016.6.1.048.
- Gillard, G.C. et al. (1979) The influence of mechanical forces on the glycosaminoglycan content of the rabbit flexor digitorum profundus tendon. *Connective Tissue Research*. 7(1): 37–46. doi: 10.3109/03008207909152351.
- Gross, J. (1959) Studies on the formation of collagen: IV. Effect of vitamin C deficiency on the neutral salt-extractible collagen of skin. *Journal of Experimental Medicine*. 109(6): 555–569.
- Hammes, H.-P. et al. (1991) Aminoguanidine treatment inhibits the development of experimental diabetic retinopathy Wycation/hyperglycemia/microvascular disease/endothelial cells/pericytes). *Proc Natl Acad Sci USA*.
- Hayashi, M. et al. (2013) The effect of lubricin on the gliding resistance of mouse intrasynovial tendon. *PLoS ONE*. 8(12): e83836. doi: 10.1371/journal.pone.0083836.
- Heinemeier, K.M. et al. (2013) Lack of tissue renewal in human adult Achilles tendon is revealed by nuclear bomb (14) C. *FASEB journal : official publication of the Federation of American Societies for Experimental Biology*. 27(5): 2074–2079. doi: 10.1096/fj.12-225599.
- Hema, G.S. et al. (2017) Optimization of process parameters for the production of collagen peptides from fish skin (*Epinephelus malabaricus*) using response surface methodology and its characterization. *Journal of Food Science and Technology*. 54(2): 488–496. doi: 10.1007/s13197-017-2490-2.
- Kapacee, Z. et al. (2008) Tension is required for fibroblast formation. *Matrix Biology*. 27(4): 371–375. doi: 10.1016/j.matbio.2007.11.006.
- Kjaer, M. (2004) Role of extracellular matrix in adaptation of tendon and skeletal muscle to mechanical loading. doi: 10.1152/physrev.00031.2003.
- Laurent, G.J., McAnulty, R.J. & Gibson, J. (1985) Changes in collagen synthesis and degradation during skeletal muscle growth. *American Journal of Physiology-Cell Physiology*. 18(2): 1–4. doi: 10.1152/ajpcell.1985.249.3.c352.
- Lind, J. (1753) *A Treatise of the Scurvy in Three Parts*.
- Lis, D.M. & Baar, K. (2019) Effects of different vitamin C-enriched collagen derivatives on collagen synthesis. *International Journal of Sport Nutrition and Exercise Metabolism*. doi: 10.1123/ijnsnem.2018-0385.
- Lundborg, G., Holm, S. & Myrhage, R. (1980) The role of the synovial fluid and tendon sheath for flexor tendon nutrition. *Scandinavian Journal of Plastic and Reconstructive Surgery and Hand Surgery*. 14(1): 99–107. doi: 10.3109/02844318009105739.
- Majima, T. et al. (2003) Stress shielding of patellar tendon: effect on small-diameter collagen fibrils in a rabbit model. *Journal of Orthopaedic Science : official journal of the Japanese Orthopaedic Association*. 8(6): 836–841. doi: 10.1007/s00776-003-0707-x.
- Manske, P.R., Bridwell, K. & Lesker, P.A. (1978) Nutrient pathways to flexor tendons of chickens using tritiated proline. *Journal of Hand Surgery. American Society for Surgery of the Hand*. 3(4): 352–357. doi: 10.1016/S0363-5023(78)80036-7.
- Manske, P.R. & Lesker, P.A. (1982) Nutrient pathways of flexor tendons in primates. *Journal of Hand Surgery. American Society for Surgery of the Hand*. 7(5): 436–444. doi: 10.1016/S0363-5023(82)80035-X.
- Marturano, J.E. et al. (2013) Characterization of mechanical and biochemical properties of developing embryonic tendon. *Proceedings of the National Academy of Sciences of the United States of America*. 110(16): 6370–6375. doi: 10.1073/pnas.1300135110.
- McAlindon, T.E. et al. (2011) Change in knee osteoarthritis cartilage detected by delayed gadolinium enhanced magnetic resonance imaging following treatment with collagen hydrolysate:

Nutrition and loading to improve fascia function

- A pilot randomized controlled trial. *Osteoarthritis and Cartilage*. 19(4): 399–405. doi: 10.1016/j.joca.2011.01.001.
- Miller, B.F. et al. (2005) Coordinated collagen and muscle protein synthesis in human patella tendon and quadriceps muscle after exercise. *Journal of Physiology*. 567(3): 1021–1033. doi: 10.1113/jphysiol.2005.093690.
- Miller, B.F. et al. (2007) Tendon collagen synthesis at rest and after exercise in women. *Journal of Applied Physiology*. 102(2): 541–546. doi: 10.1152/japplphysiol.00797.2006.
- Mussini, E., Hutton, J. & Udenfriend, S. (1967) Collagen proline hydroxylase in wound healing, granuloma formation, scurvy, and growth. *Science*. 157(3791): 927–929.
- Myer, G.D. et al. (2011) Did the NFL lockout expose the Achilles heel of competitive sports? *Journal of Orthopaedic & Sports Physical Therapy*. 41(10): 702–705. doi: 10.2519/jospt.2011.0107.
- Myrick, K.M. et al. (2019) Effects of season long participation on ACL volume in female intercollegiate soccer athletes. *Journal of Experimental Orthopaedics*. 6(1). doi: 10.1186/s40634-019-0182-8.
- Nishikimi, M. et al. (1988) Occurrence in humans and guinea pigs of the gene related to their missing enzyme l-gulonolactone oxidase. *Archives of Biochemistry and Biophysics*. 267(2): 842–846. doi: 10.1016/0003-9861(88)90093-8.
- Nugent, G.E. et al. (2006) Dynamic shear stimulation of bovine cartilage biosynthesis of proteoglycan 4. *Arthritis & Rheumatism*. 54(6): 1888–1896. doi: 10.1002/art.21831.
- Papakrivopoulou, J. et al. (2004) Differential roles of extracellular signal-regulated kinase 1/2 and p38MAPK in mechanical load-induced procollagen $\alpha 1(I)$ gene expression in cardiac fibroblasts. *Cardiovascular Research*. 61(4): 736–744. doi: 10.1016/j.cardiores.2003.12.018.
- Paxton, J.Z. et al. (2012) Optimizing an intermittent stretch paradigm using ERK1/2 phosphorylation results in increased collagen synthesis in engineered ligaments. *Tissue Engineering Part A*. 18(3–4): 277–284. doi: 10.1089/ten.tea.2011.0336.
- Paxton, J.Z., Grover, L.M. & Baar, K. (2010) Engineering an *in vitro* model of a functional ligament from bone to bone. *Tissue Engineering. Part A*. 16(11): 3515–3525. doi: 10.1089/ten.TEA.2010.0039.
- Praet, S.F.E. et al. (2019) Oral supplementation of specific collagen peptides combined with calf-strengthening exercises enhances function and reduces pain in achilles tendinopathy patients. *Nutrients*. 11(1): 1–16. doi: 10.3390/nu11010076.
- Reddy, G.K., Stehno-Bittel, L. & Enwemeka, C.S. (2002) Glycation-induced matrix stability in the rabbit Achilles tendon. *Archives of Biochemistry and Biophysics*. 399(2): 174–180. doi: 10.1006/abbi.2001.2747.
- Ricard-Blum, S. (2011) The collagen family. *Cold Spring Harbor Perspectives in Biology*. 3(1): 1–19. doi: 10.1101/cshperspect.a004978.
- Robbins, J.R., Evanko, S.P. & Vogel, K.G. (1997) Mechanical loading and TGF- β regulate proteoglycan synthesis in tendon. *Archives of Biochemistry and Biophysics*. 342(2): 203–211. doi: 10.1006/abbi.1997.0102.
- Robinson, K.A. et al. (2017) Decorin and biglycan are necessary for maintaining collagen fibril structure, fiber realignment, and mechanical properties of mature tendons. *Matrix Biology*. 64: 81–93. doi: 10.1016/j.matbio.2017.08.004.
- Sasarman, F. et al. (2016) Biosynthesis of glycosaminoglycans: associated disorders and biochemical tests. *Journal of Inherited Metabolic Disease*. 39(2): 173–188. doi: 10.1007/s10545-015-9903-z.
- Shaw, G. et al. (2017) Vitamin C-enriched gelatin supplementation before intermittent activity augments collagen synthesis. *American Journal of Clinical Nutrition*. 105(1). doi: 10.3945/ajcn.116.138594.
- Shaw, G., Serpell, B. & Baar, K. (2019) Rehabilitation and nutrition protocols for optimising return to play from traditional ACL reconstruction in elite rugby union players: A case study. *Journal of Sports Sciences*. 37(15): 1794–1803. doi: 10.1080/02640414.2019.1594571.
- Siegel, R.C. (1976) Collagen cross-linking: Synthesis of collagen cross-links in vitro with highly purified lysyl oxidase. *Journal of Biological Chemistry*. Available at: <http://www.jbc.org/> (Accessed: 1 April 2018).
- Siegel, R.C. & Fu, J.C.C. (1976) *Collagen Cross-Linking: Purification and Substrate Specificity of Lysyl Oxidase, The Journal of Biological Chemistry*. Available at: <http://www.jbc.org/> (Accessed: 1 April 2018).
- Skyhar, M.J. et al. (1985) Nutrition of the anterior cruciate ligament. Effects of continuous passive motion. *American*

Chapter 6

Journal of Sports Medicine. 13(6): 415–418. doi: 10.1177/036354658501300609.

Smeets, J.S.J. et al. (2019) Protein synthesis rates of muscle, tendon, ligament, cartilage, and bone tissue in vivo in humans. *PLOS ONE*. 14(11): e0224745. doi: 10.1371/journal.pone.0224745.

Vieira, C.P. et al. (2015) Green tea and glycine aid in the recovery of tendinitis

of the Achilles tendon of rats. *Connective Tissue Research*. 56(1): 50–58. doi: 10.3109/03008207.2014.983270.

Vieira, C.P. et al. (2015) Glycine improves biochemical and biomechanical properties following inflammation of the achilles tendon. *Anatomical Record (Hoboken, N.J. : 2007)*. 298(3): 538–545. doi: 10.1002/ar.23041.

Whiteside, L.A. & Sweeney, R.E. (1980) Nutrient pathways of the cruciate ligaments. An experimental study using the hydrogen wash-out technique. *Journal of Bone and Joint Surgery. American volume*. 62(7): 1176–1180. Available at: <http://www.ncbi.nlm.nih.gov/pubmed/7430206> (Accessed: 10 June 2019).